Run 00500 overview

Total triggers: 6241 (100.00%)

Triggers with clusters: 1362 (21.82%)

Triggers with BM hits: 1362 (21.82%)

Total clusters: 3928 (0.63 per trigger)

Avg clusters per trigger (all dets): 0.63

Beam configuration

No matching entry in beam log

Input file: SCD_RUN00500_BEAM_20250902_121516

Avg clusters per trigger (per detector)

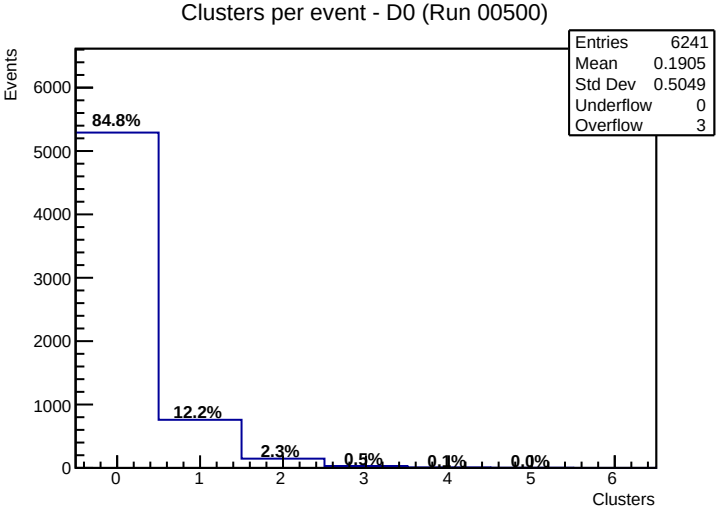
D0: 0.92 (1255/1362)

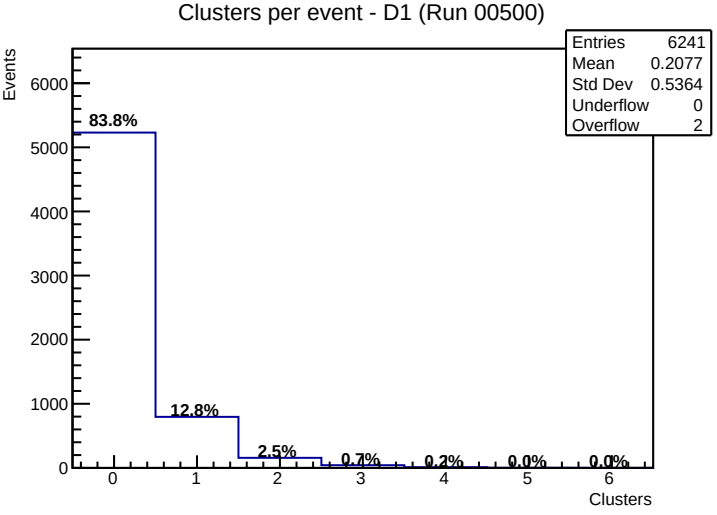
D1: 0.98 (1339/1362)

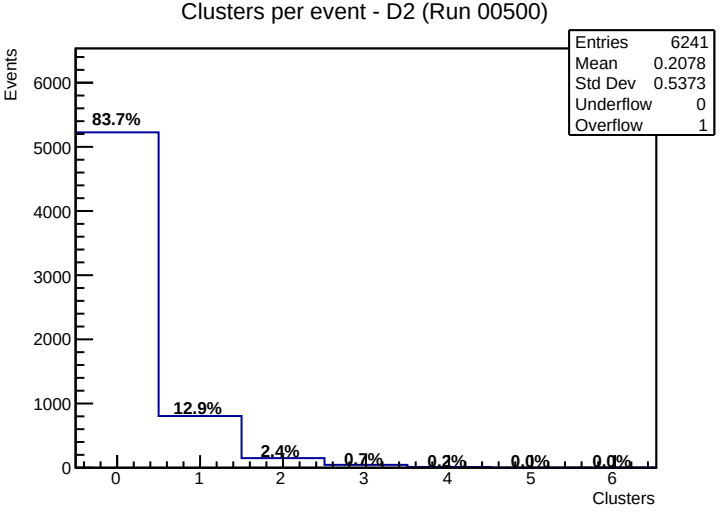
D2: 0.98 (1334/1362)

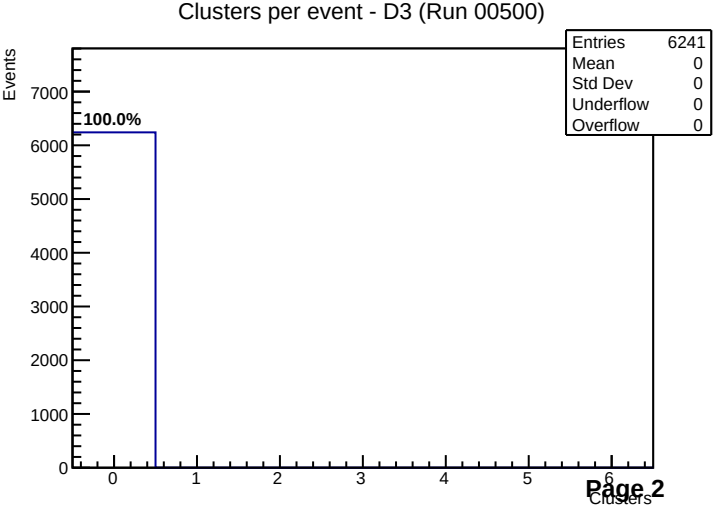
D3: 0.00 (0/1362)

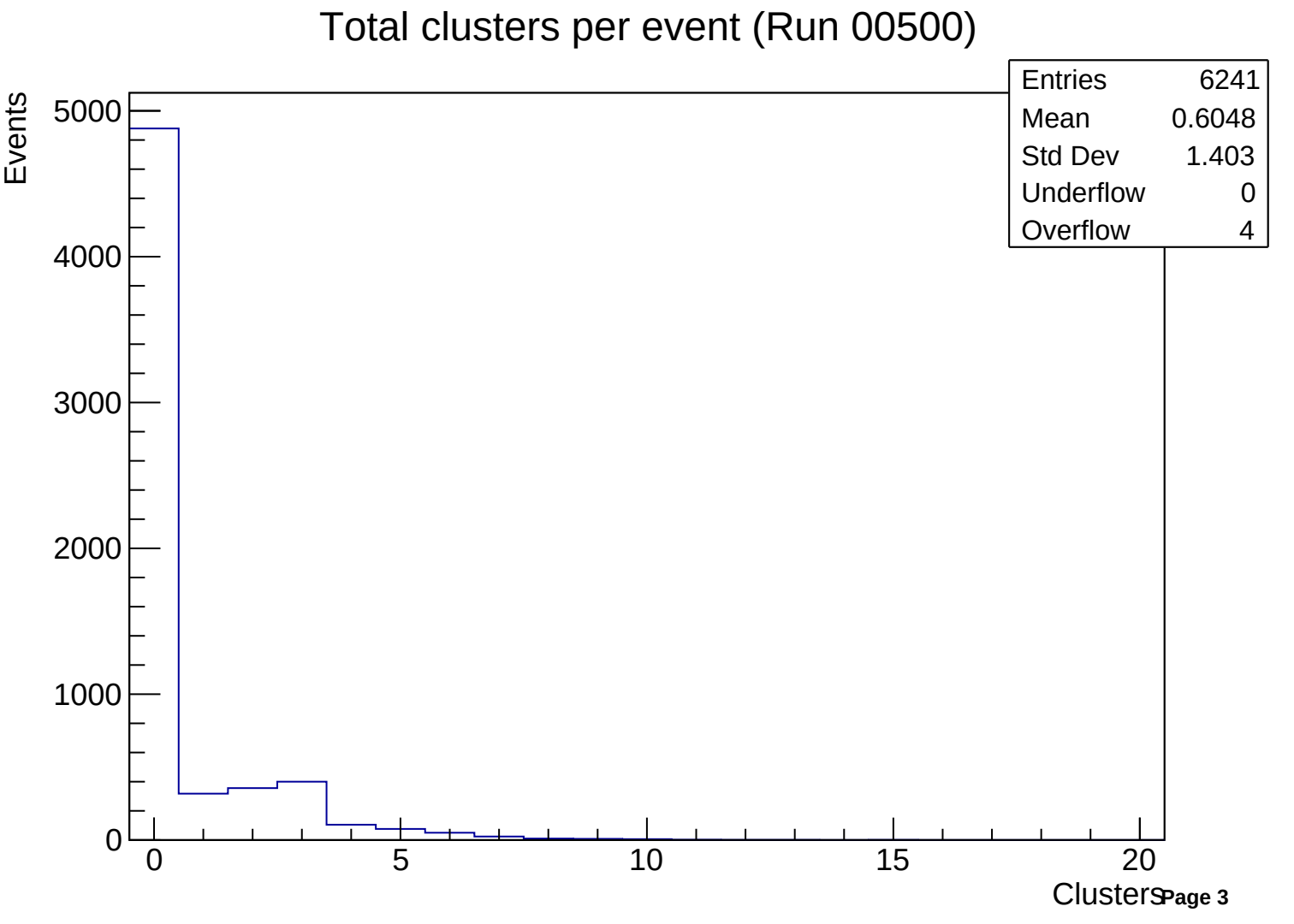
All averages use triggers with clusters (1362) as denominator.

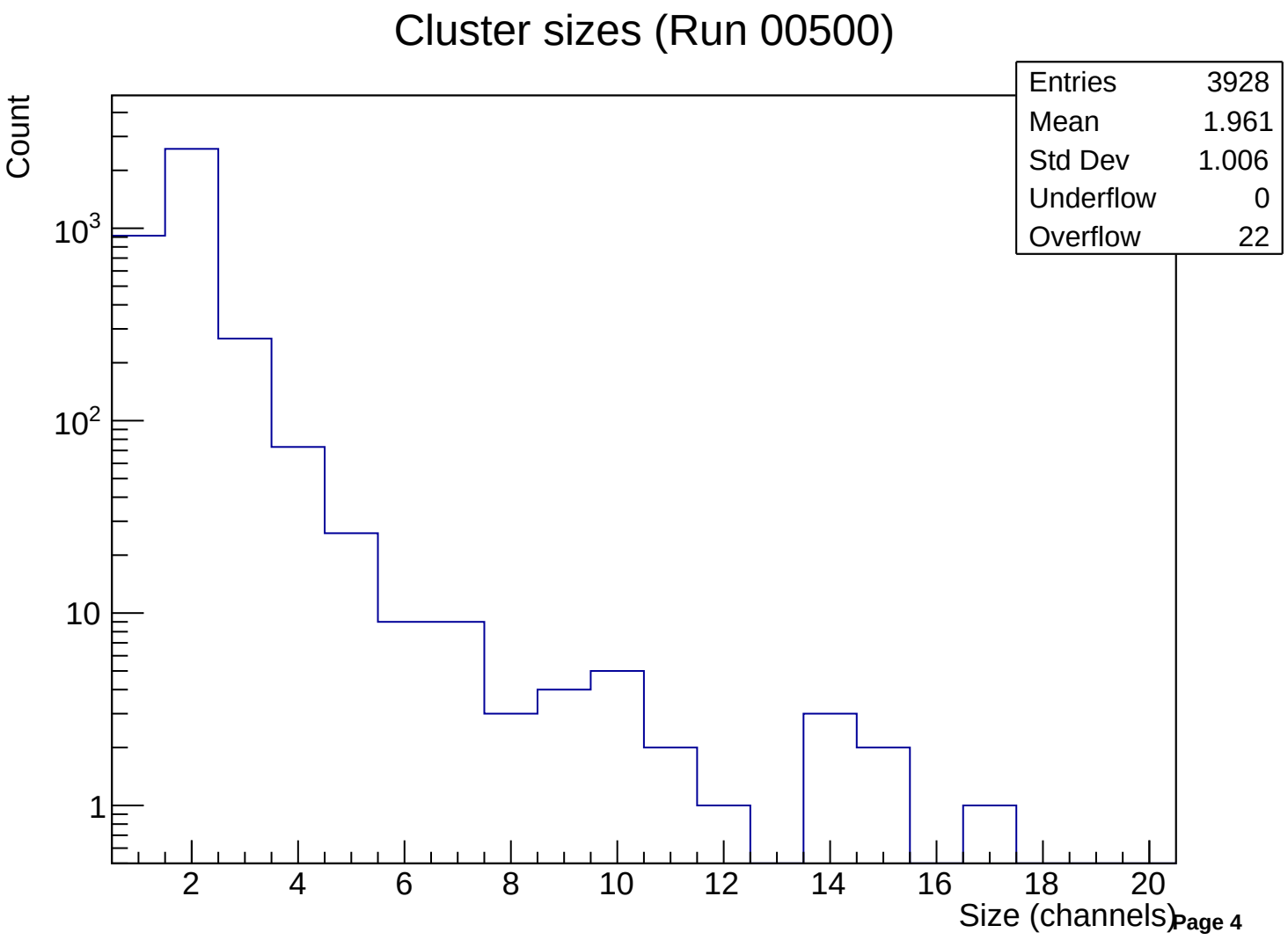

Clusters per event - D0 (Run 00500)
Entries 6241
Mean 0.1905
Std Dev 0.5049
Underflow 0
Overflow 3
84.8%
12.2%
2.3%
0.5%
0.1%
0.0%
Events
Clusters

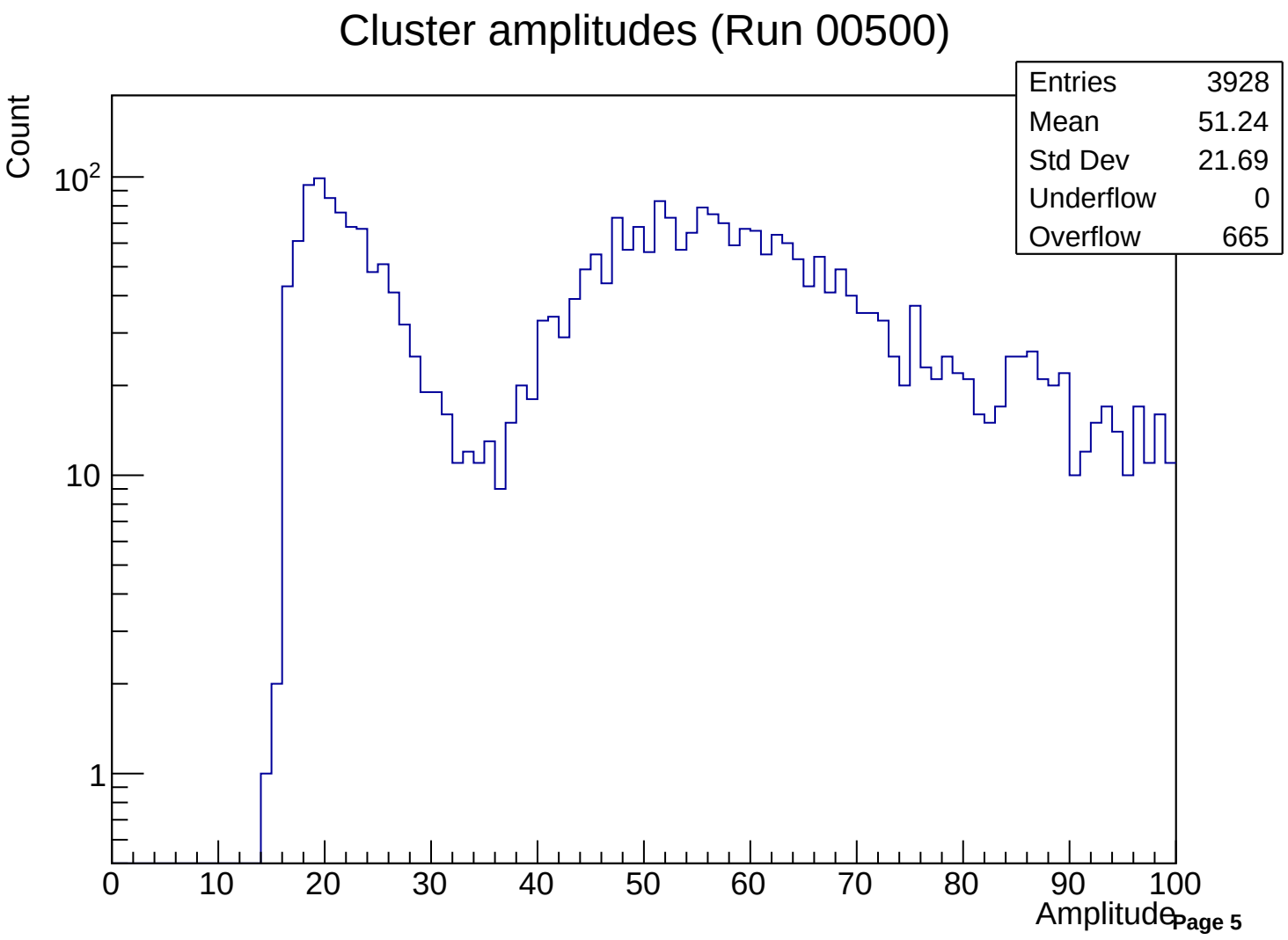

Clusters per event - D1 (Run 00500)
Entries 6241
Mean 0.2077
Std Dev 0.5364
Underflow 0
Overflow 2
83.8%
12.8%
2.5%
0.7%
0.2%
0.0%
0.0%
Events
Clusters


Clusters per event - D2 (Run 00500)
Entries 6241
Mean 0.2078
Std Dev 0.5373
Underflow 0
Overflow 1
83.7%
12.9%
2.4%
0.7%
0.2%
0.0%
0.0%
Events
Clusters

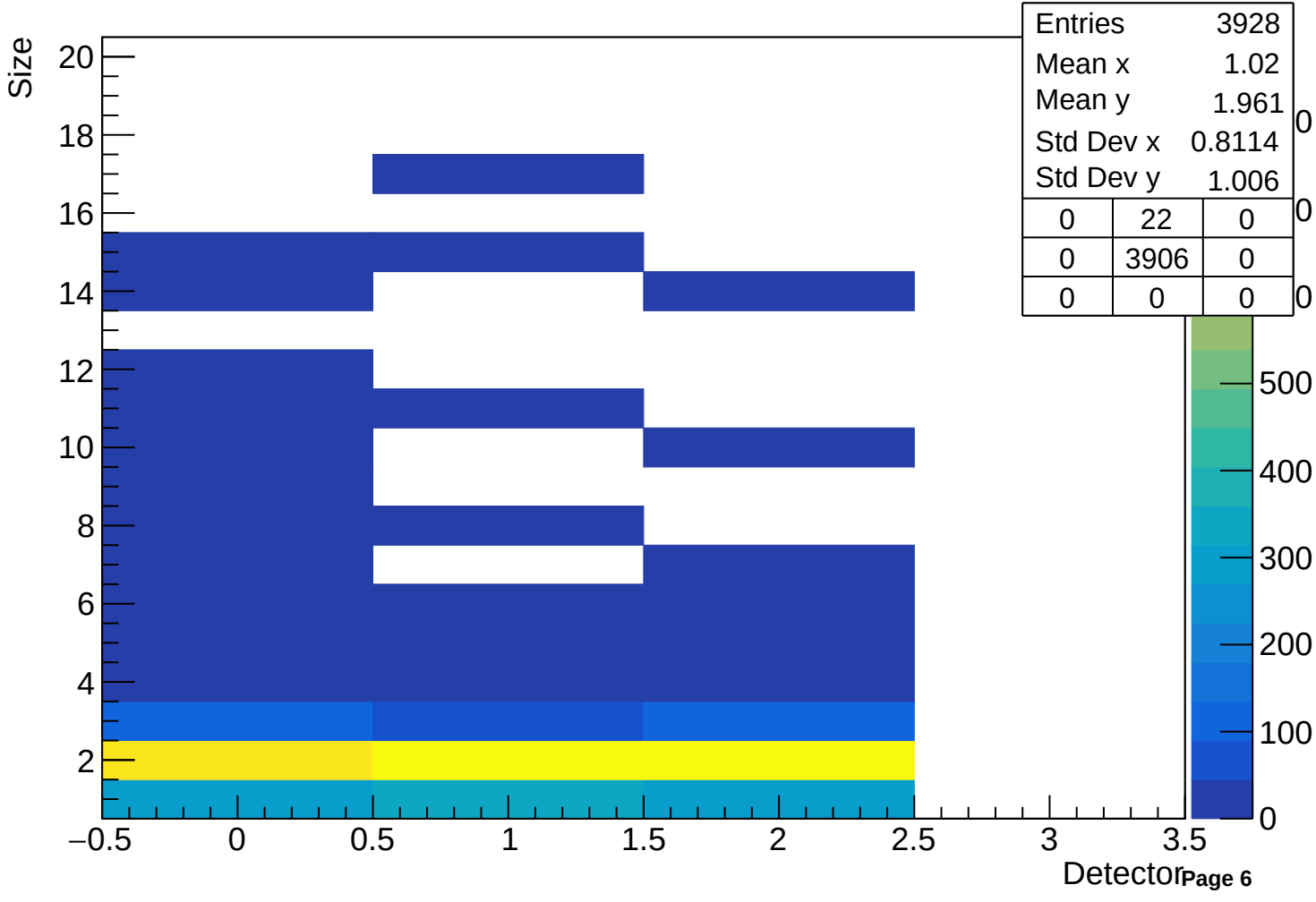

Clusters per event - D3 (Run 00500)
Entries 6241
Mean 0
Std Dev 0
Underflow 0
Overflow 0
100.0%
Events
Clusters


Total clusters per event (Run 00500)
Entries 6241
Mean 0.6048
Std Dev 1.403
Underflow 0
Overflow 4
Events
5000
4000
3000
2000
1000
0
0
5
10
15
20
Clusters

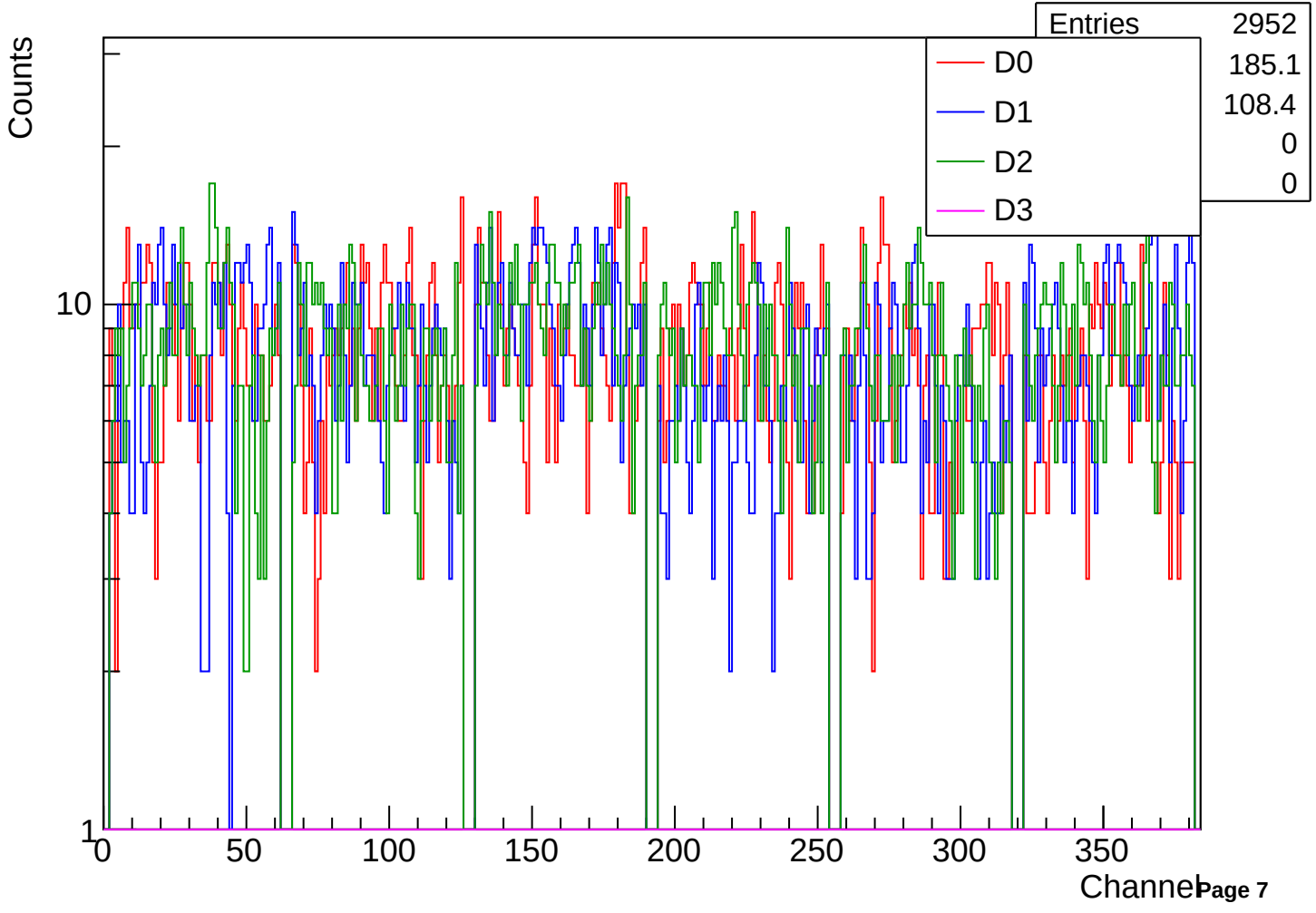

Cluster sizes (Run 00500)
Entries 3928
Mean 1.961
Std Dev 1.006
Underflow 0
Overflow 22
Count
10^3
10^2
10
1
2
4
6
8
10
12
14
16
18
20
Size (channels)

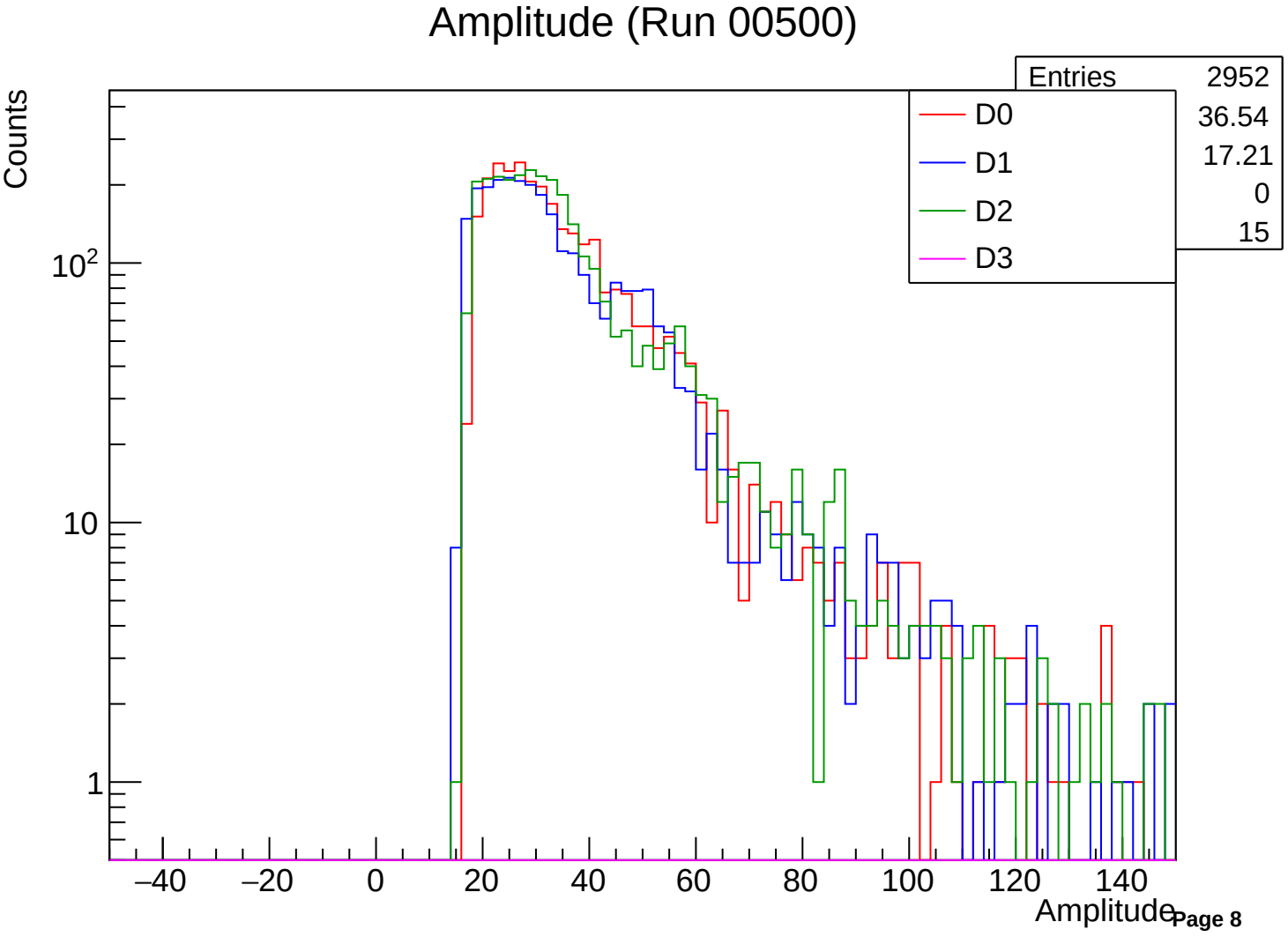

Cluster amplitudes (Run 00500)
Entries 3928
Mean 51.24
Std Dev 21.69
Underflow 0
Overflow 665
Count
10^2
10
1
0
10
20
30
40
50
60
70
80
90
100
Amplitude

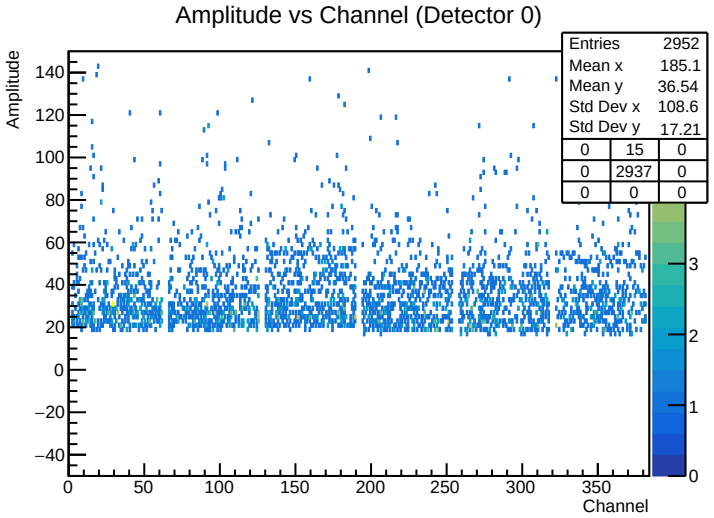
Cluster size vs detector (Run 00500)

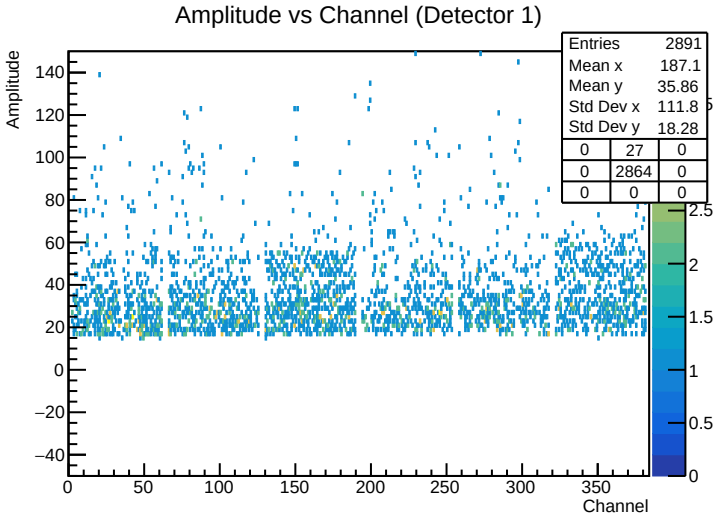


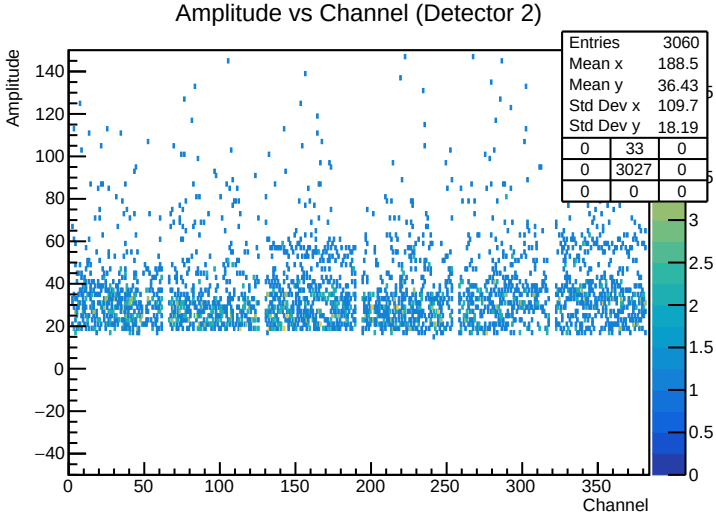
Firing channels (Run 00500)

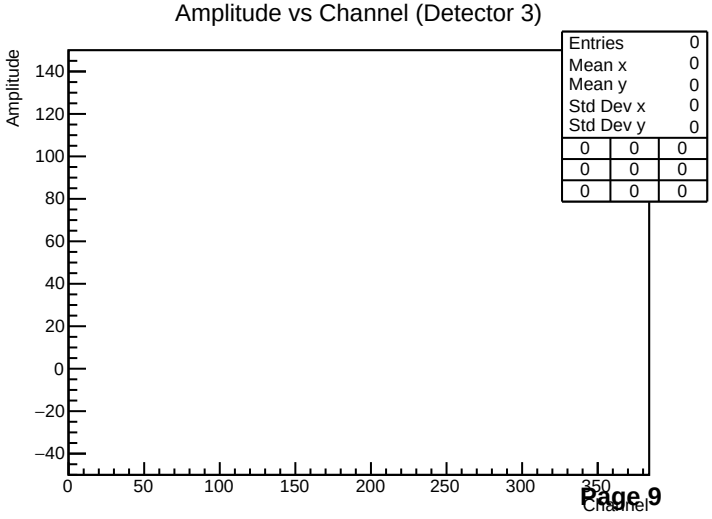



Amplitude (Run 00500)
Entries 2952
36.54
17.21
0
15
D0
D1
D2
D3
Counts
10^2
10
1
–40
–20
0
20
40
60
80
100
120
140
Amplitude

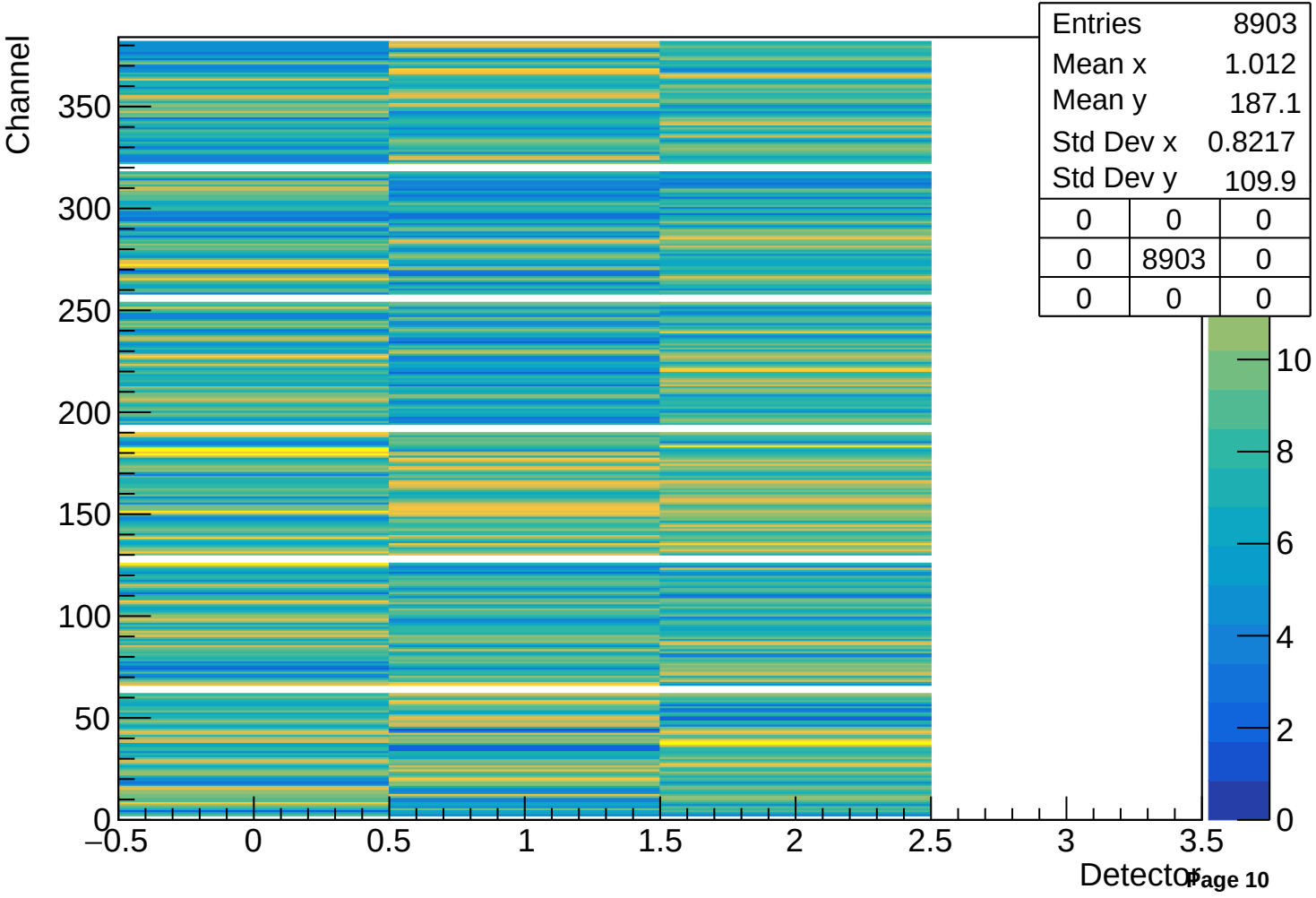

Amplitude vs Channel (Detector 0)
Entries 2952
Mean x 185.1
Mean y 36.54
Std Dev x 108.6
Std Dev y 17.21
0 15 0
0 2937 0
0 0 0
Amplitude
Channel


Amplitude vs Channel (Detector 1)
Entries 2891
Mean x 187.1
Mean y 35.86
Std Dev x 111.8
Std Dev y 18.28
0 27 0
0 2864 0
0 0 0
Amplitude
Channel

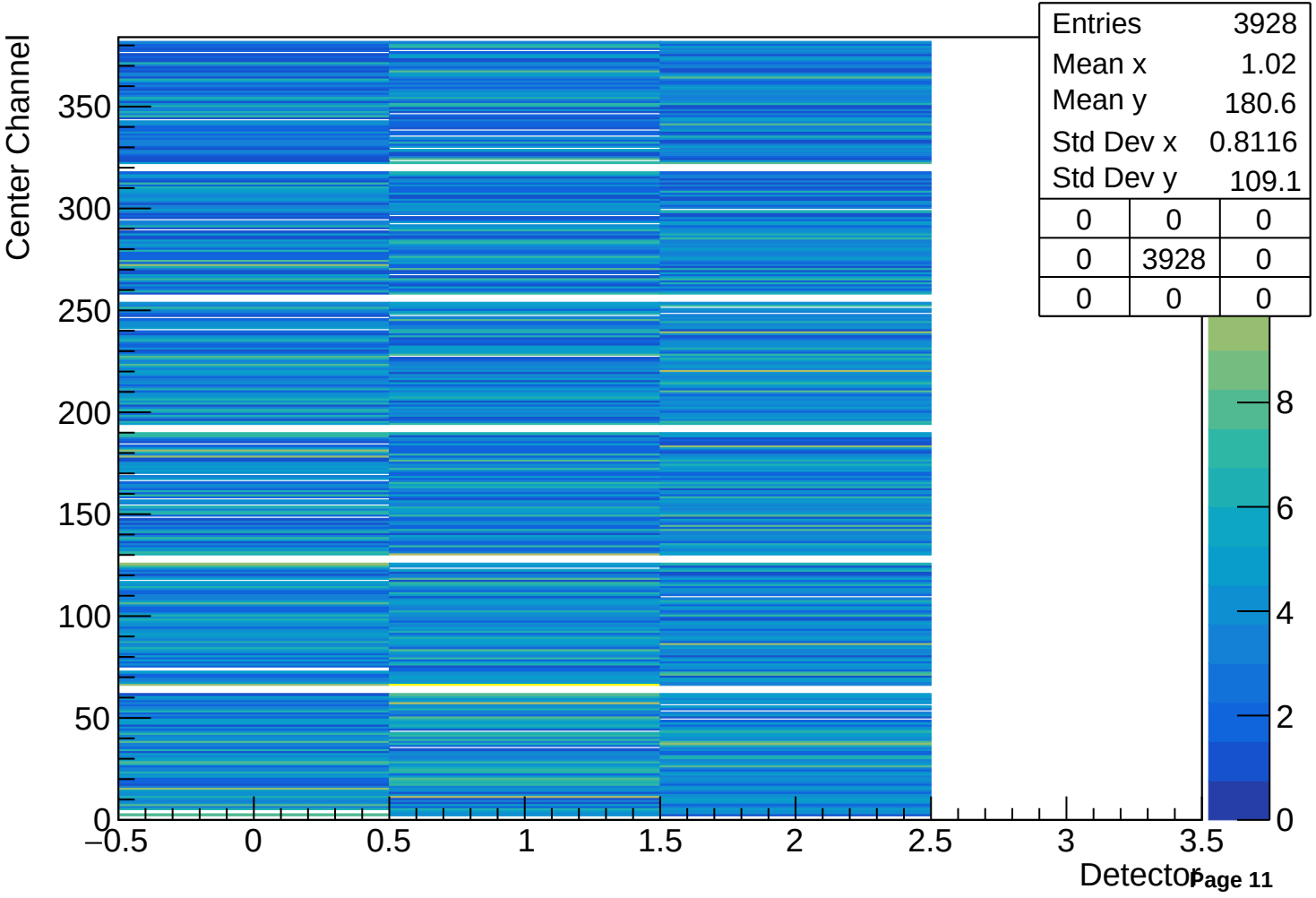

Amplitude vs Channel (Detector 2)
Entries 3060
Mean x 188.5
Mean y 36.43
Std Dev x 109.7
Std Dev y 18.19
0 33 0
0 3027 0
0 0 0
Amplitude
Channel

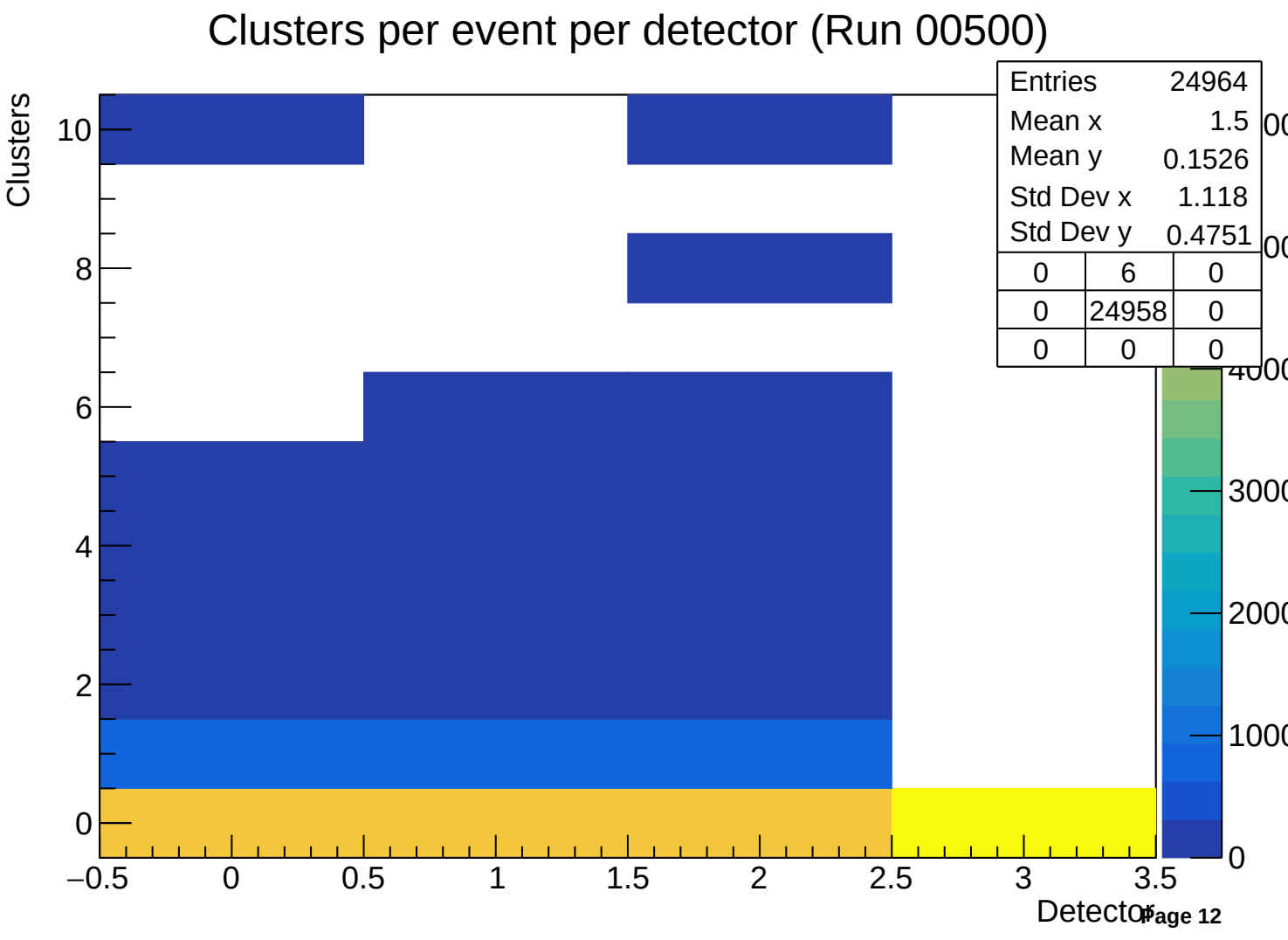

Amplitude vs Channel (Detector 3)
Entries 0
Mean x 0
Mean y 0
Std Dev x 0
Std Dev y 0
0 0 0
0 0 0
0 0 0
Amplitude
Channel

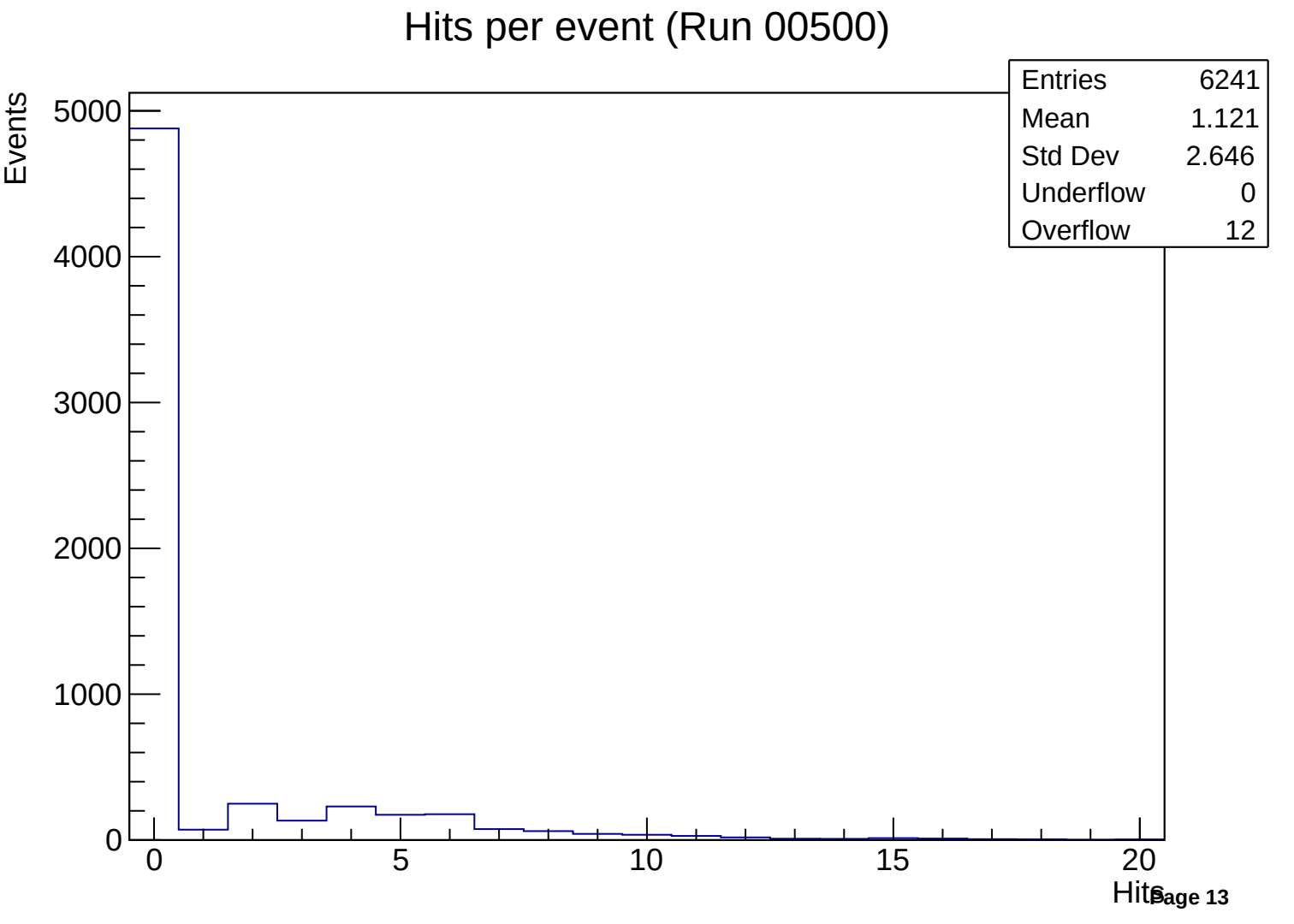
Cluster positions (Run 00500)

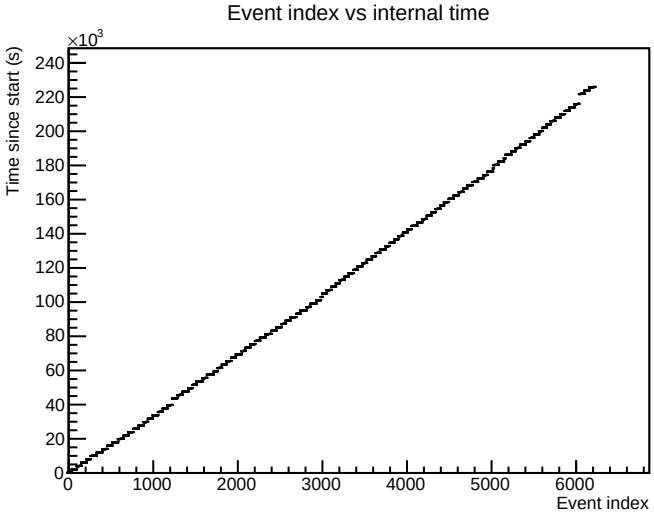


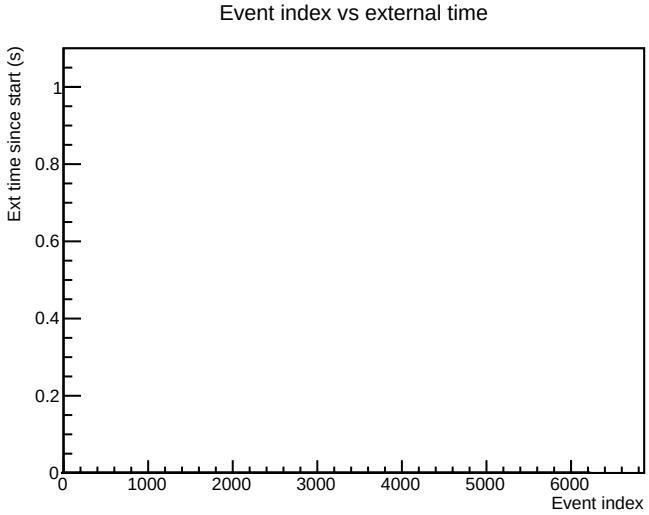
Cluster center positions (Run 00500)

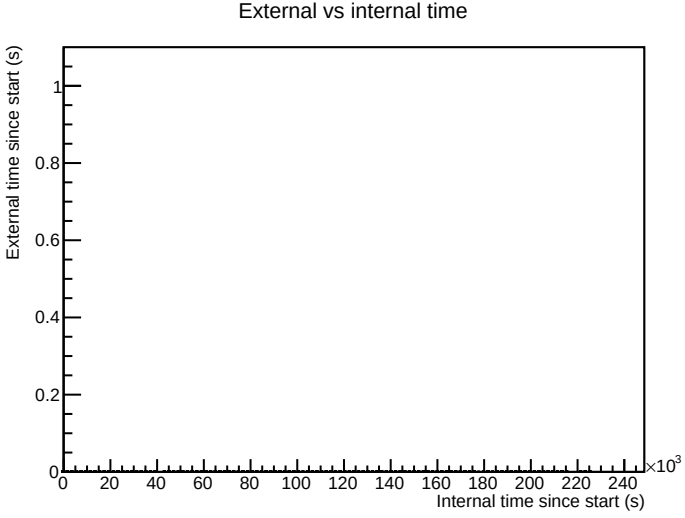


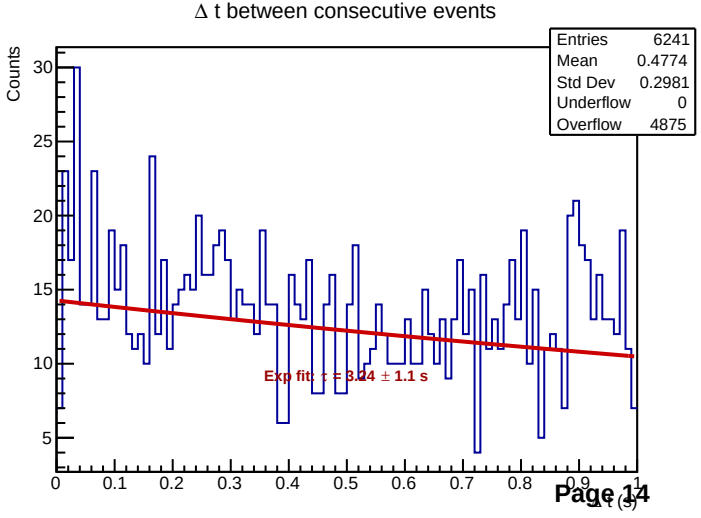

Clusters per event per detector (Run 00500)
Entries 24964
Mean x 1.5
Mean y 0.1526
Std Dev x 1.118
Std Dev y 0.4751
0 6 0
0 24958 0
0 0 0
Clusters
Detector
10
8
6
4
2
0
–0.5
0
0.5
1
1.5
2
2.5
3
3.5
4000
3000
2000
1000
0

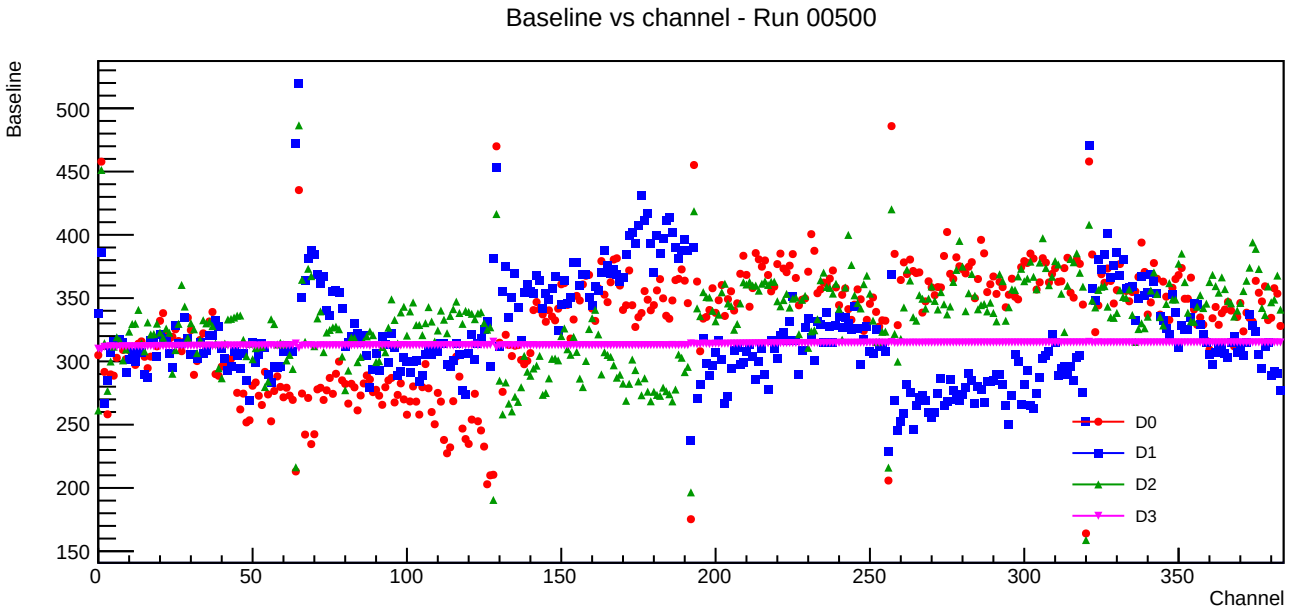

Hits per event (Run 00500)
Entries 6241
Mean 1.121
Std Dev 2.646
Underflow 0
Overflow 12
Events
5000
4000
3000
2000
1000
0
0
5
10
15
20
Hits

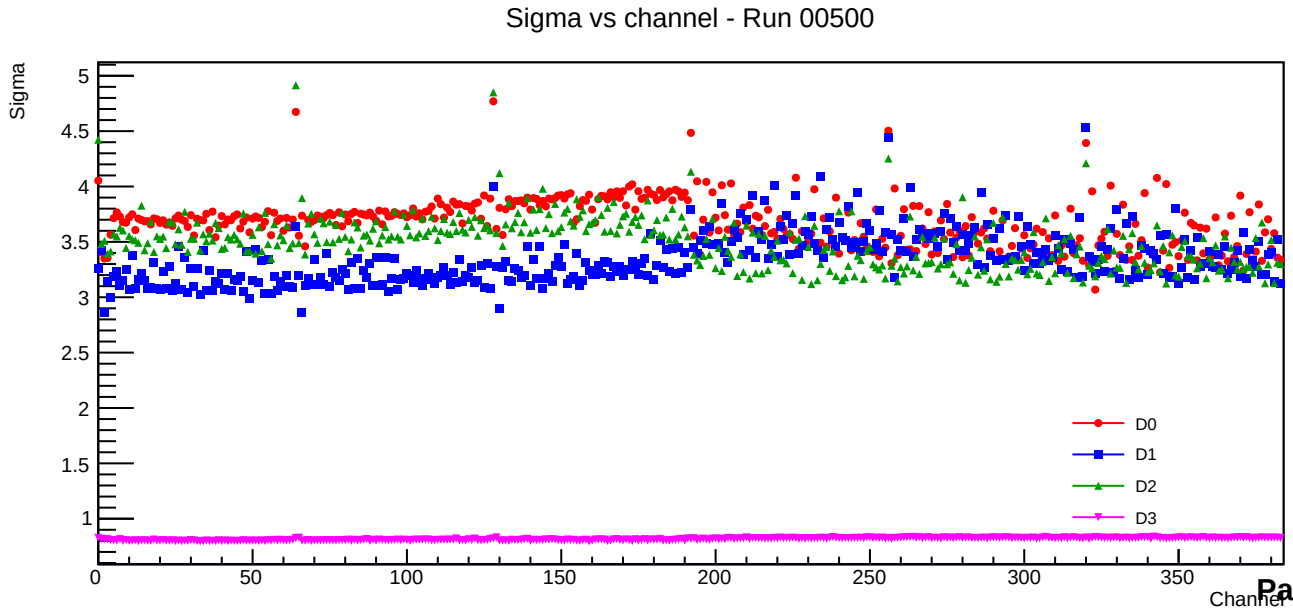

Event index vs internal time
Time since start (s)
×10^3
Event index


Event index vs external time
Ext time since start (s)
Event index


External vs internal time
External time since start (s)
×10^3
Internal time since start (s)


Δ t between consecutive events
Counts
Entries 6241
Mean 0.4774
Std Dev 0.2981
Underflow 0
Overflow 4875
Exp fit: τ = 3.24 ± 1.1 s
Δ t (s)


Baseline vs channel - Run 00500
Baseline
Channel
D0
D1
D2
D3


Sigma vs channel - Run 00500
Sigma
Channel
D0
D1
D2
D3